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1. Determine whether the following composition laws are associative, whether they are commutative and whether they have an identity:

(a) $(x, y) \mapsto x - y$ on $\mathbb{Z}$;

(b) $(x, y) \mapsto x + y + xy$ on $\mathbb{Z}$; on $\mathbb{Q} \setminus \{-1\}$.

Does any of them define a group structure? justify!

Answer: (a)
Associativity: No!

$$((x, y), z) = (x - y, z) = (x - y) - z = x - y - z,$$
$$(x, (y, z)) = (x, y - z) = x - (y - z) = x - y + z.$$

$$x - y - z \neq x - y + z \quad \text{in } \mathbb{Z}$$

Commutative: No!

$$(x, y) = x - y, \qquad (y, x) = y - x$$
$$x - y \neq y - x \quad \text{in } \mathbb{Z}$$

Identity: No!
Suppose we have the identity called e. Then it satisfies, for every $x \in \mathbb{Z}$,

$$(x, e) = x \quad (1), \qquad (e, x) = x \quad (2).$$

$$(1) \iff x - e = x \implies e = 0,$$
$$(2) \iff e - x = x \implies x = 0.$$

It is contradiction to "$\forall x \in \mathbb{Z}$".
So $\nexists$ identity.

(b)
Associativity: Yes! both on $\mathbb{Z}$ and $\mathbb{Q} \setminus \{-1\}$.

$$\begin{aligned}((x, y), z) &= (x + y + xy, z) \\ &= (x + y + xy) + z + (x + y + xy)z \\ &= x + y + z + xy + yz + zx + xyz,\end{aligned}$$
$$\begin{aligned}(x, (y, z)) &= (x, y + z + yz) \\ &= x + (y + z + yz) + x(y + z + yz) \\ &= x + y + z + xy + yz + zx + xyz.\end{aligned}$$

So $((x, y), z) = (x, (y, z))$.
Commutative: Yes! both on $\mathbb{Z}$ and $\mathbb{Q} \setminus \{-1\}$.

$$(x, y) = x + y + xy$$

$$(y, x) = y + x + yx = x + y + yx$$

$$\Downarrow \quad x, y, z \in \mathbb{Z} \text{ or } \mathbb{Q} \setminus \{-1\}$$

So $(x, y) = (y, x)$.
Identity: the identity is 0**.**
Since, for every $x \in \mathbb{Z}$ or $\mathbb{Q} \setminus \{-1\}$,

$$\begin{cases} (x, 0) = x + 0 + x \cdot 0 = x, \\ (0, x) = 0 + x + 0 \cdot x = x, \end{cases}$$

$$\Rightarrow \ (x, 0) = x = (0, x)$$

So 0 is the identity.

Only (b) on $\mathbb{Q} \setminus \{-1\}$ determine a group structure.
(a): can't because not associative.
(b): $(x, y) \mapsto x + y + xy$ is $G \times G \longrightarrow G$.
We only need inverse:

$$(x, x^{-1}) = 0 \Rightarrow x + x^{-1} + xx^{-1} = 0$$
$$\Rightarrow x^{-1} = -\frac{x}{1 + x}.$$

Inverse exist only for $x \neq -1$.
Therefore can't on $\mathbb{Z}$, only can on $\mathbb{Q} \setminus \{-1\}$.

2. Prove that a group G that has the property that $a^2 = e$ for all $a \in G$ is abelian.

Answer: proof:
Since $\forall a \in G: \quad a^2 = a \cdot a = e$,

$$\text{then } a^{-1}(a \cdot a) = a^{-1} \cdot e$$
$$\Rightarrow (a^{-1} \cdot a)a = a^{-1}$$
$$\Rightarrow a = a^{-1}$$

take $x, y \in G$

$$xy = x^{-1}y^{-1} = (yx)^{-1} = yx$$

So G is abelian.

3. Let a and b be elements of a group G. Prove that the order of ab is equal to the order of ba.

Answer: proof:
Suppose the order of ab is n.
That is: n is the minimal number s.t. $(ab)^n = e$.
Now to prove the order of ba is also n:

- To show $(ba)^n = e$:

$$\begin{aligned}(ba)^n &= (ba)(ba)\cdots(ba)\\ &= b(ab)(ab)\cdots(ab)a\\ &= b(ab)(ab)\cdots(ab)a(bb^{-1})\\ &= b(ab)(ab)\cdots(ab)(ab)b^{-1}\\ &= b(ab)^n b^{-1}\\ &= b\cdot e\cdot b^{-1}\\ &= e.\end{aligned}$$

- To show n is the smallest:
  Suppose $\exists m < n \ / \ (ba)^m = e$.

$$\begin{aligned}(ba)^m &= (ba)(ba)\cdots(ba)\\ &= b(ab)(ab)\cdots(ab)a\\ &= b(ab)(ab)\cdots(ab)a(bb^{-1})\\ &= b(ab)(ab)\cdots(ab)(ab)b^{-1}\\ &= b(ab)^m b^{-1}.\end{aligned}$$

Then

$$\begin{aligned}b(ab)^m b^{-1} &= e\\ \Rightarrow b(ab)^m b^{-1}b &= e\cdot b\\ \Rightarrow b(ab)^m &= b\\ \Rightarrow b^{-1}b(ab)^m &= b^{-1}b\\ \Rightarrow (ab)^m &= e.\end{aligned}$$

It is contradiction to order $(ab) = n$ $(m < n)$.

Therefore: order $(ba) = n$.

4. Let G be a group. Prove that the following subsets of G are subgroups.

(a) $H_1 \cap H_2$ where $H_i \leq G$.

(b) The *centraliser* of $a \in G$ defined by $C_G(a) = \{x \in G \mid ax = xa\}$.

(c) The *centre* of G defined by $Z(G) = \{x \in G \mid xy = yx,\ \forall y \in G\}$.

Answer: (a)
proof: we have the property that:
Property: $H \leq G$ iff

$$\textcircled{1}\ e \in H, \qquad \textcircled{2}\ \forall a, b \in H \Rightarrow ab \in H, \qquad \textcircled{3}\ \forall a \in H \Rightarrow a^{-1} \in H.$$

So we need to show ①②③ with $H_1 \cap H_2$.
①: since $H_1 \leq G,\ H_2 \leq G$,

$$\text{then } e \in H_1,\ e \in H_2 \quad \Rightarrow \quad e \in H_1 \cap H_2.$$

②: take $a, b \in H_1 \cap H_2$.

$$\begin{cases} a, b \in H_1 \xrightarrow{H_1 \leq G} ab \in H_1, \\ a, b \in H_2 \xrightarrow{H_2 \leq G} ab \in H_2. \end{cases}$$

Then $ab \in H_1 \cap H_2$.
③: take $x \in H_1 \cap H_2$.
In particular $x \in H_1 \ \wedge \ x \in H_2$.

$$\Rightarrow x^{-1} \in H_1, \ x^{-1} \in H_2 \quad \Rightarrow \quad x^{-1} \in H_1 \cap H_2.$$

Therefore $H_1 \cap H_2 \leq G$.

(b)
proof: the same, we need to show ①② ③ with $C_G(a)$.
①: since $e \cdot a = a = a \cdot e$
then $e \in C_G(a)$.
②: take $x, y \in C_G(a)$. Then

$$\begin{cases} xa = ax \quad (1), \\ ya = ay \quad (2). \end{cases}$$

$$\begin{aligned} (xy)a &= x(ya) = x(ay) \\ &= (xa)y \\ &= (ax)y \\ &= a(xy). \end{aligned}$$

So $(xy) \in C_G(a)$.
③: take $x \in C_G(a)$.

$$\text{then } xa = ax \quad (1)$$
$$(1) \Rightarrow a = x^{-1}ax$$

$$\begin{aligned} \text{then } ax^{-1} &= (x^{-1}ax)x^{-1} \\ &= (x^{-1}a)(xx^{-1}) \\ &= x^{-1}a. \end{aligned}$$

So $x^{-1} \in C_G(a)$.
Therefore $C_G(a) \leq G$.

(c)
①: Since $e \cdot y = y = y \cdot e \quad \forall y \in G$
then $e \in Z(G)$.
②: take $a, b \in Z(G)$.
Then

$$\begin{cases} ax = xa \quad \forall x \in G \quad (1), \\ by = yb \quad \forall y \in G \quad (2). \end{cases}$$

$$\begin{aligned} (ab)z = a(bz) = a(zb) &= (az)b \\ &\overset{(1)}{=} (za)b \\ &= z(ab). \end{aligned}$$

Then $ab \in Z(G)$.
③: take $a \in Z(G)$.
Then $ax = xa \quad \forall x \in G$.

$$\begin{aligned} a^{-1}x &= a^{-1}(axa^{-1}) \\ &= (a^{-1}a)(xa^{-1}) \\ &= xa^{-1} \quad \forall x \in G. \end{aligned}$$

So $a^{-1} \in Z(G)$.
Therefore $Z(G) \leq G$.

5. Prove that the inverse of a permutation matrix is its transpose.

Answer: Def: permutation matrix is a $n \times n$ matrix that every row, column has only one "1".
proof: Suppose we have a permutation matrix P.
To show $P^{-1} = P^T$ we just need to show that $PP^T = I = P^TP$.
① $PP^T = I$:
Suppose $P \in M_{n\times n}$,

$$P = (P_{ij}) = \begin{bmatrix} P_{11} & P_{12} & \cdots & P_{1n} \\ P_{21} & P_{22} & \cdots & P_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ P_{n1} & P_{n2} & \cdots & P_{nn} \end{bmatrix}, \qquad P^T = \begin{bmatrix} P_{11} & P_{21} & \cdots & P_{n1} \\ P_{12} & P_{22} & \cdots & P_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ P_{1n} & P_{2n} & \cdots & P_{nn} \end{bmatrix}.$$

Then

$$(PP^T)_{ij} = P_{i1}P_{j1} + P_{i2}P_{j2} + \cdots + P_{in}P_{jn}.$$

If $i \neq j$:

$$(PP^T)_{ij} = P_{i1}P_{j1} + P_{i2}P_{j2} + \cdots + P_{in}P_{jn} = 0.$$

If $i = j$:

$$\begin{aligned} (PP^T)_{ij} &= P_{i1}P_{i1} + P_{i2}P_{i2} + \cdots + P_{in}P_{in} \\ &= P_{i1}^2 + P_{i2}^2 + \cdots + P_{in}^2 \quad (One of them = 1, other = 0) \\ &= 1. \end{aligned}$$

So $PP^T = I$.
Similarly: $P^TP = I$.
Therefore: $P^{-1} = P^T$.